

**SEMINAR SUMMARY:
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NUMERICAL SOLUTIONS OF BIOLOGICALLY BASED PROBLEMS
USING DIFFERENTIAL EQUATIONS**

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In general across all the projects we explore below in this summary we have a general framework around how one of these projects is structured. Specifically the goal is to collaborate with biologist to understand research questions and the current state of the field. From this collaboration, our goal is to develop mathematical models that accommodate using numerical techniques for solving them. We use the biological data to parameterize the model and/or run a sweep of the parameter space. Though it is worth noting that there is a natural trade-off between mathematical complexity of the model and the solvability of the model.

The projects that we examined specifically were:

- Wound healing and inflammation
- Wound healing in coral
- Gut health and inflammation
- Nematocyst firing
- Phage therapy
- Sickle Cell Pain Episode Prediction

Our summaries on projects 5. and 6. were generally short because those both were described in the last five-minutes of class.

1. WOUND HEALING AND INFLAMMATION.

For this project the original data came from an experiment that the original biologist did on himself (with medical assistance). Specifically he had a “PTFE” (or vortex) tube inserted under his skin for a period of time to create the wound in question as a mechanism for creating a standardized wound. Not surprisingly, there were few other participants in the clinical trial. That said, because it was biologist himself doing the experiment the richness of the data is interesting. For example, that there were even mounted slides from the experiment that showed the PTFE tube matrix along with fibroblasts laying down collagen.

The model in question was a partial differential equation model based on the variables m for the density of damaged the tissue, n for the activated inflammatory cells, f for the

fibroblasts, and c for new collagen deposition. We have four model equations, starting with the first for phagocytosis:

$$\frac{\partial m}{\partial t} = -\mu_{mn}mn$$

The second equation represented diffusion and chemotaxis and deactivation, given by:

$$\frac{\partial n}{\partial t} = \nabla \cdot [D_n \nabla n - n \chi_n \nabla m] - \mu_{mn}mn$$

The third equation represented diffusion/motility and chemotaxis and cell division and cell death

$$\frac{\partial f}{\partial t} = \nabla \cdot [D_f \nabla f - f \chi_f \nabla n] + s_f(n)f \left(2 - \frac{f}{F_0}\right) - \mu_f f$$

And the fourth model equation represents deposition by f and collateral damage

$$\frac{\partial c}{\partial t} = k_c f(C_0 - c) - \mu_{cn}cn.$$

We were then presented some plots that represented collagen and fibroblast levels versus days with modeled data and real data plotted on the graphs.

2. WOUND HEALING IN CORAL.

The goal in this project is to consider coral in place of human tissue. Biologically we know that the damage processes (laceration, avulsion, fracture), inflammation processes, and physiological process for tissue regrowth behave in a similar fashion to humans. We noted that it was a convenient wound healing mechanism to study because the results could indeed be produced with coral in the lab as well as naturally occurring coral. We ask questions like whether the wound is regrowing or propagating. There were two specified aims of the project, namely cellular level-wound healing versus tissue level-energy focused

In both of the aims of the project we began examining whether coral bleaching had any effect on the speed and mechanics of wound healing. Though it is known that glycerol production as a photosynthetic byproduct transferred to mucus from the symbiotic algae yielded energy transfer to the calicoblastic cells for skeletal growth.

Potential future research projects out of this body of work could be examining multiple stressors, other coral species and linked cellular and tissue level models.

3. GUT HEALTH AND INFLAMMATION.

So it's known that gut health has a considerable amount to do with body health from general inflammation, to mental health, to even levels of atherosclerosis. Now gut inflammation is regulated by an enzyme called alkaline phosphatase (specifically intestinal alkaline phosphatase or IAP). And, the gram-negative bacteria in the colon produce lipopolysaccharides (LPS) which are inflammatory agents. So the interplay between IAP and LPS regulates how the gut performs and even the structural integrity of wall of the bowel.

If I understand the slides correctly our goal here is to write a mathematical model that effectively encompasses the metabolic pathway for the breakdown of LPS by IAP, which was given in a diagram in one of the slides. The equations that arise from this metabolic pathway follow as:

$$\begin{aligned}
\frac{dIAP}{dt} &= \alpha S_I - \rho_I IAP * LPS_G - \mu_I I \\
\frac{dP}{dt} &= \theta_I \left(\frac{LPS_G}{X_{ad} + LPS_G} \right) - \theta_R \frac{P}{P + K_R} \\
\frac{dLPS_G}{dt} &= \beta S_l - \rho_L IAP * LPS_G - P(LPS_G - LPS_B) \\
\frac{dLPS_B}{dt} &= DP(LPS_G - LPS_B) - \mu_L LPS_B
\end{aligned}$$

After getting these equations we saw a variety of plots. The first plots were actual animal data taken from lab mice. We looked at the differences between a “chow” diet and a “western” diet over a multiple week period and how inflammation occurred in the animal. Further we saw how changes in diet could affect inflammation. Lastly, we were presented with a variety of numerical plots of data generated from the model comparing a healthy diet, a continued “western” diet a partially healthy diet and then those with stimulus where there was a stimulus to affect the production of IAP in a positive manner.

4. NEMATOCYST FIRING.

This problem I personally found more interesting than the rest, as we had a display of a single cell organism firing off an organelle, that we call a nematocyst, into another single cell organism, algae, for the purpose of capture. More generally nematocysts are employed by cnidarians (hydra, jellyfish, corals) and dinoflagellates (*Polykrikos kofoiidi*), noting that the dinoflagellate mechanism seem to have more parts involved. Morphogenesis of these organelles is mostly understood, but the mechanics of firing, on the other hand, is not. The firing is so fast that it is quite difficult to capture and analyze.

The mathematical model that we employ is that of an accelerating rod that is transposed into the prey at velocity V . We have the variables L as the length of the rod, h as the distance between the prey and the rod, and D as the diameter of the prey. The model follows from the non-dimensional NavierStokes equation

$$\begin{aligned}
\frac{\partial \mathbf{v}'}{\partial t'} + \mathbf{v}' \cdot \nabla' \mathbf{v}' &= -\nabla' p' + \frac{1}{\text{Re}} \nabla'^2 \mathbf{v}' \\
\text{Re} &= \frac{\rho L V}{\mu}
\end{aligned}$$

And, note that the Navier-Stokes equation can be put into this dimensionless form using the following definitions:

$$\begin{aligned} \mathbf{x}' &= \frac{\mathbf{x}}{L} & p' &= \frac{p}{\rho V^2} \\ \mathbf{v}' &= \frac{\mathbf{v}}{V} & t' &= t \frac{V}{L} \\ \nabla' &= L \nabla \end{aligned}$$

where L is the characteristic length and V is the characteristic velocity.

We then went over the code that would be written beginning with the mesh generation that we might create using finite element analysis or the finite difference method. Then the object would be to move the boundary around to accommodate for muscles, biochemistry, preferred curvature and other items. We notice that the model is fairly elastic with two different angles upon which we look at it: the first being a fiber based model where things hook together using beams and springs, the second being a continuum model using finite elements.

We then considered a boundary layer approach where we model everything with spring mass dampers with a multitude of little springs to represent the boundary layer of the fluid. This lead to some interesting modeled results show as a video with a green background where the nematocysts never actually punctured the prey. We then looked at the difference across Reynolds numbers, namely for $\text{Re} = 0.02$ versus $\text{Re} = 200$. And we were mainly concerned with the ability of the nematocyst to reach the prey. Which, we plotted in various other plots.

Ideas on future direction follow as: determine how nematocyst is ejected from the cell (applied force or pressure), and to study the mechanics of how the capsule is triggered.

5. PHAGE THERAPY.

So we didn't really get into the mathematical model behind the phage therapy, but we did get into what the basis of the problem is biologically, and it's quite interesting. The idea is to develop unique bacteriophages that are specifically tailored to the bacteria's transmembrane receptors. Note, bacteriophages essentially behave with a bacterial cell in the fashion that a virus would with a human cell. So the goal with the production of these bacteriophages is to administer them to patients with particularly officious bacterial infections. Furthermore, utilizing the technique for patients that are dealing with antibiotic resistant bacteria or are particularly immunocompromised.

We were very briefly shown two models one with delay differential equations (DDE) and one with ordinary differential equations (ODE). The ODE model was shown to fit the data better than the DDE model. Though, we can still gain insight from both models. For

example, the DDE model tends to predict an earlier emergence of resistance as compared with the ODE model. However, in the long run the ODE model will be used for our project

6. SICKLE CELL PAIN EPISODE PREDICTION.

Similarly to the “Phage therapy” project we got an abbreviated presentation on the mathematical modeling of sickle cell anemia. The goal behind this project essentially is to see how pain works. We are using ODEs to model the cell dynamics of the hemoglobin distortions. The mechanism of pain is vascular occlusion and our goal is to properly model this. Further we’re looking at inflammation generally associated with the anemia as well as how sleep affects recovery from an episode.